Part III

From Short Rates to the Market Model: How BGM Came to Be

CHAPTER 9

Short-Rate Models and Their Limits

9.1 A different way to build a model

Parts I and II worked entirely with market-observable objects: discount factors, forward rates, swap rates, quoted volatilities. An older and historically prior tradition in rates modelling instead starts from a single, theoretical, instantaneous quantity — the **short rate** $r(t)$, informally "the interest rate earned over the next infinitesimal instant" — and builds an entire curve and every derivative price out of assumed dynamics for that one process. This chapter is a brief tour of that tradition, not because any of its specific models will reappear later — they will not — but because understanding precisely what they get right and where they break down is what makes the BGM alternative, built in the following two chapters, feel like a necessity rather than an arbitrary modelling choice.

Key result: pricing from the short rate

Given assumed risk-neutral dynamics for $r(t)$, every discount factor and every derivative price follows, in principle, from a single expectation:

$$P(t,T) = \mathbb{E}_t\left[\exp\left(-\int_t^T r(s)\,\mathrm{d}s\right)\right], \qquad V(t) = \mathbb{E}_t\left[\exp\left(-\int_t^T r(s)\,\mathrm{d}s\right) H_T\right]$$

for a payoff H_T at time T. The entire modelling problem reduces to choosing dynamics for $r(t)$ that are both economically sensible and tractable enough that these expectations can actually be computed.

9.2 Vasicek, CIR, and Hull–White

The earliest tractable short-rate models are all variations on **mean-reverting** dynamics — $r(t)$ is pulled back toward a long-run level, with some form of Brownian noise superimposed.

Key result: three classical short-rate models

Vasicek (1977): $\mathrm{d}r = a(b-r)\,\mathrm{d}t + \sigma\,\mathrm{d}W$. Gaussian, mean-reverting, analytically tractable (bond prices and options are closed-form), but r can go negative with positive probability — once considered a flaw, later, ironically, a feature during the negative-rate years.

Cox–Ingersoll–Ross, CIR (1985): $\mathrm{d}r = a(b-r)\,\mathrm{d}t + \sigma\sqrt{r}\,\mathrm{d}W$. The $\sqrt{r}$ term forces $r \geq 0$ (under a parameter condition, the Feller condition), fixing Vasicek's negativity but at the cost of a less convenient, non-Gaussian distribution.

Hull–White (1990): $\mathrm{d}r = (\theta(t) - ar)\,\mathrm{d}t + \sigma\,\mathrm{d}W$ — Vasicek with a time-dependent level $\theta(t)$, chosen specifically so the model can be made to fit *today's* entire observed yield curve exactly, not just a single point on it.

The progression across these three models is itself instructive: each model fixes a specific, named problem with its predecessor, a pattern that repeats throughout this entire part of the book.

9.3 The problem these models share

All short-rate models, however sophisticated, share one structural limitation that no amount of extra parameters fixes: they describe the evolution of a single, unobservable, instantaneous number, from which an *entire curve* at every future date must somehow be derived. Two consequences follow, both fatal for the purpose this book cares about.

PITFALL

The calibration mismatch. The market does not trade the short rate. It trades caps, floors and swaptions, each written on a specific observable forward rate or swap rate over a specific period. A one-factor short-rate model, having only one source of randomness, forces every point on the curve to move in lockstep (perfectly correlated), which is observably false — the market prices caplets and swaptions of different tenors at genuinely different, imperfectly correlated volatilities, and a single-factor model simply cannot reproduce that pattern, however its one free volatility parameter is tuned. Multi-factor short-rate models exist and partially address this, but at a steep cost in tractability, and the fundamental mismatch between "the model's state variable" and "the market's quoted instrument" remains.

PITFALL

The translation problem. Even where a short-rate model can be calibrated reasonably well, doing so requires translating market-observed cap and swaption *volatilities* into short-rate model *volatility parameters* through a chain of non-trivial, model-specific formulas — there is no direct, one-to-one correspondence between "the market quotes 80 bp normal vol on this caplet" and "therefore $\sigma = 0.012$ in Hull–White." Every short-rate model requires its own bespoke calibration machinery to bridge this gap, and the resulting fit is typically good near the specific instruments used to calibrate and degrades away from them.

ON THE DESK

Short-rate models like Hull–White have not disappeared from trading desks — a one-factor Hull–White model, extended and recalibrated frequently, remains a common, fast, and well-understood tool for Bermudan-style callable products where a Markovian, low-dimensional model is a genuine computational advantage over a full BGM simulation. What changed is the *default* choice for anything where getting the initial cap and swaption smile right, cleanly and directly, matters more than getting maximum computational speed. That default shifted decisively toward market models once BGM made it possible to calibrate directly to what actually trades, which is exactly the story of the next two chapters.

9.4 The road not fully closed: HJM

Before market models existed, one framework had already partially solved the translation problem — by discarding the short rate as the sole state variable and modelling the *entire forward curve* directly. That framework, Heath–Jarrow–Morton, is the subject of the next chapter, and it is the direct intellectual ancestor of BGM: BGM is, in a precise sense that

Chapter 11 makes exact, what you get when you apply the HJM idea to *discrete, market-observable* forward rates instead of the theoretical instantaneous ones HJM itself uses.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. State the defining SDE of the Vasicek, CIR and Hull–White models, and identify what specific problem each one fixes relative to its predecessor.
2. Explain why a single-factor short-rate model cannot reproduce imperfectly correlated volatilities across different points of the curve.
3. Explain, in one sentence, why calibrating a short-rate model to market cap/swaption quotes is a harder problem than it sounds, even when the model itself is tractable.